6.22 FURTHER PURE MATHEMATICS WITH TECHNOLOGY, FPT (4798) A2

Objectives

To build on and extend students' knowledge of Pure Mathematics through using technology to:

- perform mathematical processes quickly and accurately;
- observe the effect of changing parameters displayed in different representations, which is useful for aiding generalisation;
- engage in investigative approaches to problem solving.

Assessment

Examination (72 marks)
Up 2 hours
All questions are compulsory.
Three questions each worth about 24 marks.

Candidates require access to a computer with a computer algebra system (CAS), graphing software, spreadsheet program and programming language throughout the examination.

Assumed Knowledge

Candidates are expected to know the content of *C1*, *C2*, *C3*, *C4*, *FP1* and the Complex Numbers, Polar curves, Power Series and Hyperbolic Functions sections of *FP2*.

Calculators

In the MEI Structured Mathematics specification, no calculator is allowed in the examination for *C1*. For all other units, including this one, a graphical calculator is allowed.

Computer Software

Centres must complete a form seeking approval from OCR for the software they intend to use for the examination. The form is available on the website (www.ocr.org.uk) or by requesting it from ocrmathematics@ocr.org.uk. Centres can obtain advice from MEI about suitable software (www.mei.org.uk).

Use of CAS

Candidates may use CAS for any of the algebraic processes in this unit such as solving, factorising, expanding, differentiating, integrating and finding limits. However, candidates should take care to give sufficient evidence when answering "show that" questions.

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Specification	**Ref.**	**Competence Statements**

INVESTIGATION OF CURVES

In this topic students develop skills associated with curves. They learn to look for and recognise important properties of curves, making appropriate use of graphing software and CAS. They are expected to be able to generalise their findings; at times this will require analytical skills.

Examination questions will use a variety of curves but candidates will not be expected to know their particular properties. Instead the questions will test candidates' ability to select and apply the skills to investigate them. It is, however, anticipated that while studying this option, students will meet a wide selection of curves including curves expressed as cartesian equations, parametric equations and polar curves. Candidates will be expected to use CAS to solve equations and evaluate derivatives.

Curves.	FPTC1	Know and use the vocabulary associated with curves.
Graphing Software	2	Know how to plot a family of curves in graphing software.
Properties of Curves.	3	Be able to find, describe and generalise properties of curves.
	4	Be able to determine asymptotes.
	5	Be able to find the gradient of the tangent to a curve at a point.
	6	Be able to find and work with equations of chords, tangents and normals.
	7	Be able to use the limit of f(x) as $x \to a$ or $x \to \infty$ to deduce and investigate properties of a curve.
	8	Be able to identify cusps by examining the behaviour nearby.

INVESTIGATION OF CURVES

FPTC1	e.g. asymptote, cusp, loop, node; terms relating to symmetry. Students should know the names and shapes of conics.		
2	Students will be expected to be able to use sliders for parameters.		
3	Generalisation may involve exploratory use of graphing software. Any algebraic work may involve the exploratory use of CAS. Curves may be given in cartesian, polar or parametric form.		
4	Including oblique asymptotes and asymptotic approach to curves.		
5	The general form of the formula for the gradient of a tangent to a polar curve will be given if required.		
6	Derivatives can be found with CAS.		
7	Limits can be found with CAS.		
8	Examining the limit of the gradient as the curve approaches the cusp from both directions.		

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FUNCTIONS OF COMPLEX VARIABLES

Students will be expected to use the complex number capabilities of CAS. The use of spreadsheets is expected for iterations and limits. i will be used for $\sqrt{-1}$ throughout.

Functions	FPTj1	Understand what a function of a complex variable is and be able to identify the real and imaginary parts of a function.
Equations	2	Be able to the interpret the solution of an equation given by CAS.
	3	Understand that the roots of a polynomial with complex coefficients will be complex numbers.
	4	Be able to find and use the roots of a polynomial with complex coefficients.
Differentiation	5	Be able to differentiate polynomials, sin z, sinh z, cos z, cosh z, e^z.
Indefinite integration	6	Be able to find the indefinite integrals of polynomials, sin z, sinh z, cos z, cosh z, and e^z.
Limits	7	Be able to use the limit of a function of a complex variable.
	8	Be able to construct a spreadsheet to demonstrate how a function behaves as it tends to a limit.
Trigonometric, hyperbolic and exponential functions	9	Be able to use the relationships between functions of complex variables.
Iterations	10	Be able to perform iterations of a function of a complex variable

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Ref.	**Notes**	**Notation**	**Exclusions**

FUNCTIONS OF COMPLEX VARIABLES

1	e.g. $\text{real}((x+y\text{i})^3) = x^3 - 3xy^2$ To include polynomials, rational functions, sin z, sinh z, cos z, cosh z, e^z		
2	To include the representation of solutions on the Argand diagram.		
3	Students are expected to know that a polynomial of degree n has n complex roots.		Proof of the fundamental theorem of algebra
4	Students will be expected to solve a polynomial with complex coefficients using CAS.		
5			Tests for differentiability. The Cauchy-Riemann equations.
6	e.g. using indefinite integration to solve $\text{f}\,'(z) = \cos z$.		
7	Limits will be restricted to $\lim\limits_{h\to 0} \text{f}((x+h)+\text{i}y)$ or $\lim\limits_{h\to 0} \text{f}(x+\text{i}(y+h))$. CAS may be used.		
8			
9	The following may be used: $\sin(x+\text{i}y) = \sin x \cosh y + \text{i}\cos x \sinh y$ $\sinh(x+\text{i}y) = \sinh x \cos y + \text{i}\cosh x \sin y$ $\cos(x+\text{i}y) = \cos x \cosh y - \text{i}\sin x \sinh y$ $\cosh(x+\text{i}y) = \cosh x \cos y + \text{i}\sinh x \sin y$ $e^{x+\text{i}y} = e^x e^{\text{i}y}$ $= e^x \cos y + \text{i}e^x \sin y$ Students are expected to be able to express these relationships in terms of Maclaurin series.		
10	To include the geometrical representation.		

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NUMBER THEORY

The assessment of this topic will be based on the assumption that candidates have a suitable programming language.

Examination questions will feature programs that produce the solutions to problems in Number Theory. Candidates may be expected to write their own programs as well as understanding a program and suggesting limitations and refinements to it.

Specification	Ref.	Competence Statements
Programming	FPTT1	Be able to write programs to solve number theory problems.
	2	Be able to identify the limitations of a short program and suggest refinements to it.
Factorisation	3	Know the unique factorisation of natural numbers.
	4	Be able to solve problems using modular arithmetic.
	5	Know Fermat's little theorem.
Diophantine Equations	6	Be able to solve linear Diophantine equations and use solutions to solve related problems.
	7	Be able to find Pythagorean triples and use related equations.
	8	Be able to solve Pell's equation and use solutions to solve related problems.
	9	Be able to solve other Diophantine equations and use solutions to solve related problems.
Gaussian integers	10	Know the definition of Gaussian integers and be able to find and use Gaussian primes.

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NUMBER THEORY

FPTT1	IF, WHILE, FOR, LOOP, GOTO, local/global variables, inputting/outputting variables, basic lists/arrays (1D only). Problems will be taken from statements 3-10.		
2	Students will be expected to discuss, informally, the efficiency of a program and offer a small number of changes to it.		
3			Proof of the fundamental theorem of arithmetic
4		$16 \equiv 2 (\mathrm{mod}\, 7)$	
5	Use of the terms co-prime and GCD/HCF.		
6			
7	Related equations may include different indices, such as $x^2 + y^2 = z^3$.		
8			
9	All the required information will be given in the question.		
10			

7 Further Information and Training for Teachers

The specifications are supported by a complete package provided by MEI and OCR.

On-line Support

- Web-based resources covering all the units in this specification (www.mei.org.uk; www.ocr.org.uk).

Teaching Materials

- Textbooks, one for each unit in the specifications.
- Students' Handbook.

INSET and Teacher Support

- One-day INSET courses provided by both MEI and OCR.
- The MEI annual three-day conference.
- MEI branch meetings.
- Help from both MEI and OCR at the end of the telephone.
- Regular newsletters from MEI.

Examinations

- Specimen examination papers and mark schemes.
- Past examination papers and mark schemes.
- Examiners' reports.
- Practice papers for new units.

Coursework

- A bank of coursework resource materials.
- Exemplar marked coursework.
- Reports to Centres from coursework moderators.

Addresses

The MEI Office	OCR
Monckton House, Epsom Centre	1 Hills Road
White Horse Business Park	Cambridge
Trowbridge, Wilts. BA14 0XG	CB1 2EU

Appendix A: Mathematical Formulae

This appendix lists formulae that candidates are expected to remember and that may not be included in formulae booklets.

Beside each formula is the unit in which it is first encountered.

Quadratic Equations

$ax^2+bx+c=0$ has roots $\dfrac{-b\pm\sqrt{b^2-4ac}}{2a}$ C1

Laws of Logarithms

$\log_a x+\log_a y\equiv\log_a(xy)$

$\log_a x-\log_a y\equiv\log_a\left(\dfrac{x}{y}\right)$ C2

$k\log_a x\equiv\log_a\left(x^k\right)$

Trigonometry

In the triangle ABC

$$\frac{a}{\sin A}=\frac{b}{\sin B}=\frac{c}{\sin C}$$ C2

Area $=\dfrac{1}{2}ab\sin C$ C2

$\cos^2 A+\sin^2 A\equiv 1$ C2

$\sec^2 A\equiv 1+\tan^2 A$ C4

$\operatorname{cosec}^2 A\equiv 1+\cot^2 A$ C4

$\sin 2A\equiv 2\sin A\cos A$ C4

$\cos 2A\equiv\cos^2 A-\sin^2 A$ C4

$\tan 2A\equiv\dfrac{2\tan A}{1-\tan^2 A}$ C4

Differentiation

Function	**Derivative**	
x^n	nx^{n-1}	C2
$\sin kx$	$k\cos kx$	C4
$\cos kx$	$-k\sin kx$	C4
e^{kx}	ke^{kx}	C3
$\ln x$	$\frac{1}{x}$	C3
$f(x)+g(x)$	$f'(x)+g'(x)$	C2
$f(x)g(x)$	$f'(x)g(x)+f(x)g'(x)$	C3
$f(g(x))$	$f'(g(x))g'(x)$	C3

Integration

Function	**Integral**	
x^n	$\frac{1}{n+1}x^{n+1}+c,\ n\neq -1$	C2
$\cos kx$	$\frac{1}{k}\sin kx+c$	C4
$\sin kx$	$-\frac{1}{k}\cos kx+c$	C4
e^{kx}	$\frac{1}{k}e^{kx}+c$	C3
$\frac{1}{x}$	$\ln\lvert x\rvert+c,\ x\neq 0$	C2
$f'(x)+g'(x)$	$f(x)+g(x)+c$	C2
$f'(g(x))g'(x)$	$f(g(x))+c$	C3

Area

Area under a curve $=\int_a^b y\,dx\ \ (y\geq 0)$ C2

Vectors

$\begin{pmatrix} x \\ y \\ z \end{pmatrix} \bullet \begin{pmatrix} a \\ b \\ c \end{pmatrix} = xa+yb+zc$ C4

Appendix B: Mathematical Notation

1 Set Notation

$\in$	is an element of
$\notin$	is not an element of
$\{x_1,\ x_2,\ldots\}$	the set with elements $x_1,\ x_2,\ \ldots$
$\{x : \ldots\}$	the set of all x such that $\ldots$
$\mathrm{n}(A)$	the number of elements in set A
$\varnothing$	the empty set
$\mathcal{E}$	the universal set
A'	the complement of the set A
$\mathbb{N}$	the set of natural numbers, $\{1,\ 2,\ 3,\ \ldots\}$
$\mathbb{Z}$	the set of integers, $\{0,\ \pm 1,\ \pm 2,\ \pm 3,\ \ldots\}$
$\mathbb{Z}^+$	the set of positive integers, $\{1,\ 2,\ 3,\ \ldots\}$
$\mathbb{Z}_n$	the set of integers modulo n, $\{0,\ 1,\ 2,\ \ldots\ ,\ n-1\}$
$\mathbb{Q}$	the set of rational numbers, $\left\{\dfrac{p}{q} : p \in \mathbb{Z},\ q \in \mathbb{Z}^+\right\}$
$\mathbb{Q}^+$	the set of positive rational numbers, $\{x \in \mathbb{Q} : x > 0\}$
$\mathbb{Q}_0^+$	set of positive rational numbers and zero, $\{x \in \mathbb{Q} : x \geq 0\}$
$\mathbb{R}$	the set of real numbers
$\mathbb{R}^+$	the set of positive real numbers, $\{x \in \mathbb{R} : x > 0\}$
$\mathbb{R}_0^+$	the set of positive real numbers and zero, $\{x \in \mathbb{R} : x \geq 0\}$
$\mathbb{C}$	the set of complex numbers
$(x,\ y)$	the ordered pair x, y
$A \times B$	the cartesian product of sets A and B, i.e. $A \times B = \{(a,\ b) : a \in A,\ b \in B\}$
$\subseteq$	is a subset of
$\subset$	is a proper subset of
$\cup$	union
$\cap$	intersection
$[a,\ b]$	the closed interval $\{x \in \mathbb{R} : a \leq x \leq b\}$

$[a, b)$	the interval $\{x \in \mathbb{R} : a \leq x < b\}$
$(a, b]$	the interval $\{x \in \mathbb{R} : a < x \leq b\}$
(a, b)	the open interval $\{x \in \mathbb{R} : a < x < b\}$
$y R x$	y is related to x by the relation R
$y \sim x$	y is equivalent to x, in the context of some equivalence relation

2 Miscellaneous Symbols

$=$	is equal to
$\neq$	is not equal to
$\equiv$	is identical to or is congruent to
$\approx$	is approximately equal to
$\cong$	is isomorphic to
$\propto$	is proportional to
$<$	is less than
$\leqslant$	is less than or equal to, is not greater than
$>$	is greater than
$\geqslant$	is greater than or equal to, is not less than
∞	infinity
$p \wedge q$	p and q
$p \vee q$	p or q (or both)
$\sim p$	not p
$p \Rightarrow q$	p implies q (if p then q)
$p \Leftarrow q$	p is implied by q (if q then p)
$p \Leftrightarrow q$	p implies and is implied by q (p is equivalent to q)
$\exists$	there exists
$\forall$	for all

3 Operations

$a + b$	a plus b
$a - b$	a minus b
$a \times b,\ ab,\ a.b$	a multiplied by b
$a \div b,\ \frac{a}{b},\ a/b$	a divided by b

$\sum_{i=1}^{n} a_i$	$a_1 + a_2 + \ldots + a_n$
$\prod_{i=1}^{n} a_i$	$a_1 \times a_2 \times \ldots \times a_n$
$\sqrt{a}$	the positive square root of a
$\lvert a \rvert$	the modulus of a
$n!$	n factorial
$\binom{n}{r}$, ${}^nC_r$	the binomial coefficient $\frac{n!}{r!(n-r)!}$ for $n \in \mathbb{Z}^+$ or $\frac{n(n-1)\ldots(n-r+1)}{r!}$ for $n \in \mathbb{Q}$

4 Functions

$\mathrm{f}(x)$	the value of the function f at x
$\mathrm{f}: A \to B$	f is a function under which each element of set A has an image in set B
$\mathrm{f}: x \mapsto y$	the function f maps the element x to the element y
f^{-1}	the inverse function of the function f
gf	the composite function of f and g which is defined by $\mathrm{gf}(x) = \mathrm{g}(\mathrm{f}(x))$
$\lim_{x \to a} \mathrm{f}(x)$	the limit of $\mathrm{f}(x)$ as x tends to a
Δx, δx	an increment of x
$\frac{\mathrm{d}y}{\mathrm{d}x}$	the derivative of y with respect to x
$\frac{\mathrm{d}^n y}{\mathrm{d}x^n}$	the nth derivative of y with respect to x
$\mathrm{f}'(x)$, $\mathrm{f}''(x)$, $\ldots$, $\mathrm{f}^{(n)}(x)$	the first, second, ..., nth derivatives of $\mathrm{f}(x)$ with respect to x
$\int y \, \mathrm{d}x$	the indefinite integral of y with respect to x
$\int_a^b y \, \mathrm{d}x$	the definite integral of y with respect to x between the limits $x = a$ and $x = b$
$\frac{\partial V}{\partial x}$	the partial derivative of V with respect to x
$\dot{x}$, $\ddot{x}$, $\ldots$	the first, second, ... derivatives of x with respect to t

5 Exponential and Logarithmic Functions

e	base of natural logarithms
e^x, $\exp x$	exponential function of x
$\log_a x$	logarithm to the base a of x
$\ln x$, $\log_e x$	natural logarithm of x
$\lg x$, $\log_{10} x$	logarithm of x to base 10

6 Circular and Hyperbolic Functions

$\left.\begin{array}{l}\sin,\ \cos,\ \tan,\\ \operatorname{cosec},\ \sec,\ \cot\end{array}\right\}$	the circular functions
$\left.\begin{array}{l}\arcsin,\ \arccos,\ \arctan,\\ \operatorname{arcosec},\ \operatorname{arcsec},\ \operatorname{arcot}\end{array}\right\}$	the inverse circular functions
$\left.\begin{array}{l}\sinh,\ \cosh,\ \tanh,\\ \operatorname{cosech},\ \operatorname{sech},\ \coth\end{array}\right\}$	the hyperbolic functions
$\left.\begin{array}{l}\operatorname{arcsinh},\ \operatorname{arcosh},\ \operatorname{arctanh},\\ \operatorname{arcosech},\ \operatorname{arcsech},\ \operatorname{arcoth}\end{array}\right\}$	the inverse hyperbolic functions

7 Complex Numbers

i,j	square root of -1
z	a complex number, $z = x + \mathrm{j}y = r(\cos\theta + \mathrm{j}\sin\theta)$
$\operatorname{Re} z$	the real part of z, $\operatorname{Re} z = x$
$\operatorname{Im} z$	the imaginary part of z, $\operatorname{Im} z = y$
$\lvert z \rvert$	the modulus of z, $\lvert z \rvert = \sqrt{x^2 + y^2}$
$\arg z$	the argument of z, $\arg z = \theta$, $-\pi < \theta \le \pi$
z^*	the complex conjugate of z, $x - \mathrm{j}y$

8 Matrices

$\mathbf{M}$	a matrix $\mathbf{M}$
$\mathbf{M}^{-1}$	the inverse of the matrix $\mathbf{M}$
$\mathbf{M}^{\mathrm{T}}$	the transpose of the matrix $\mathbf{M}$
$\det \mathbf{M}$ or $\lvert\mathbf{M}\rvert$	the determinant of the square matrix $\mathbf{M}$

9 Vectors

$\mathbf{a}$	the vector $\mathbf{a}$
$\overrightarrow{AB}$	the vector represented in magnitude and direction by the directed line segment AB
$\hat{\mathbf{a}}$	a unit vector in the direction of $\mathbf{a}$
$\mathbf{i}$, $\mathbf{j}$, $\mathbf{k}$	unit vectors in the directions of the cartesian coordinate axes
$\lvert\mathbf{a}\rvert$, a	the magnitude of $\mathbf{a}$
$\lvert\overrightarrow{AB}\rvert$, AB	the magnitude of $\overrightarrow{AB}$
$\mathbf{a.b}$	the scalar product of $\mathbf{a}$ and $\mathbf{b}$
$\mathbf{a}\times\mathbf{b}$	the vector product of $\mathbf{a}$ and $\mathbf{b}$

10 Probability and Statistics

A, B, C, etc.	events
$A\cup B$	union of the events A and B
$A\cap B$	intersection of the events A and B
$\mathrm{P}(A)$	probability of the event A
A'	complement of the event A
$\mathrm{P}(A\mid B)$	probability of the event A conditional on the event B
X, Y, R, etc.	random variables
x, y, r, etc.	values of the random variables X, Y, R etc
x_1, x_2, $\ldots$	observations
f_1, f_2, $\ldots$	frequencies with which the observations x_1, x_2, $\ldots$ occur
$\mathrm{p}(x)$	probability function $\mathrm{P}(X=x)$ of the discrete random variable X
p_1, p_2, $\ldots$	probabilities of the values x_1, x_2, $\ldots$ of the discrete random variable X
$\mathrm{f}(x)$, $\mathrm{g}(x)$, $\ldots$	the value of the probability density function of a continuous random variable X
$\mathrm{F}(x)$, $\mathrm{G}(x)$, $\ldots$	the value of the (cumulative) distribution function $\mathrm{P}(X\leqslant x)$ of a continuous random variable X
$\mathrm{E}(X)$	expectation of the random variable X
$\mathrm{E}(\mathrm{g}(X))$	expectation of $\mathrm{g}(X)$
$\mathrm{Var}(X)$	variance of the random variable X
$\mathrm{G}(t)$	probability generating function for a random variable which takes the values $0, 1, 2, \ldots$

$\mathrm{B}(n,\ p)$	binomial distribution with parameters n and p
$\mathrm{N}(\mu,\ \sigma^2)$	normal distribution with mean μ and variance σ^2
μ	population mean
σ^2	population variance
σ	population standard deviation
$\overline{x},\ m$	sample mean
$s^2,\ \hat{\sigma}^2$	unbiased estimate of population variance from a sample, $s^2 = \frac{1}{n-1}\sum(x_i - \overline{x})^2$
ϕ	probability density function of the standardised normal variable with distribution $\mathrm{N}(0,\ 1)$
Φ	corresponding cumulative distribution function
ρ	product moment correlation coefficient for a population
r	product moment correlation coefficient for a sample
$\mathrm{Cov}(X,\ Y)$	covariance of X and Y